Auteur: Sabawoon Enayat
Studierichting: Informatica
Oefeningenreeks 1C nr. 48.1

Opgave: Bereken $(1+i)^n - (1-i)^n$, met $n = 2025$

$(1+i)^{2025} - (1-i)^{2025}$

$z_1 = 1 + i$

$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\Rightarrow \tan\theta = \dfrac{b}{a} = \dfrac{1}{1} = 1 \Rightarrow \theta = \mathrm{Bgtan}(1) = \dfrac{\pi}{4}$

$\Rightarrow z_1 = r \operatorname{cis}\theta = \sqrt{2}\operatorname{cis}\left(\dfrac{\pi}{4}\right)$

$z_2 = 1 - i$

$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2}$

$\Rightarrow \tan\theta = \dfrac{b}{a} = -\dfrac{1}{1} = -1 \Rightarrow \theta = \mathrm{Bgtan}(-1) = -\dfrac{\pi}{4}$

$\Rightarrow z_2 = r \operatorname{cis}\theta = \sqrt{2}\operatorname{cis}\left(-\dfrac{\pi}{4}\right)$

$\overset{\text{De Moivre}}{\Rightarrow} (r \operatorname{cis}\theta)^n = r^n \operatorname{cis}(n\theta)$

$\left(\sqrt{2}^{2025} \operatorname{cis}\left(2025\dfrac{\pi}{4}\right)\right) - \left(\sqrt{2}^{2025} \operatorname{cis}\left(-2025\dfrac{\pi}{4}\right)\right)$

$= \sqrt{2}^{2025}\left(\operatorname{cis}\left(2025\dfrac{\pi}{4}\right) - \operatorname{cis}\left(-2025\dfrac{\pi}{4}\right)\right)$

$= \sqrt{2^{2024} \cdot 2}\left(\operatorname{cis}\left(2024\dfrac{\pi}{4} + \dfrac{\pi}{4}\right) - \operatorname{cis}\left(-2024\dfrac{\pi}{4} - \dfrac{\pi}{4}\right)\right)$

$= 2^{1012} \cdot \sqrt{2}\left(\operatorname{cis}\left(\dfrac{\pi}{4}\right) - \operatorname{cis}\left(-\dfrac{\pi}{4}\right)\right)$

$= 2^{1012} \cdot \sqrt{2}\left(\cos\left(\dfrac{\pi}{4}\right) + i \cdot \sin\left(\dfrac{\pi}{4}\right) - \cos\left(-\dfrac{\pi}{4}\right) - i \cdot \sin\left(-\dfrac{\pi}{4}\right)\right)$

$= 2^{1012} \cdot \sqrt{2}\left(\dfrac{\sqrt{2}}{2} + i \cdot \dfrac{\sqrt{2}}{2} - \dfrac{\sqrt{2}}{2} + i \cdot \dfrac{\sqrt{2}}{2}\right)$

$= 2^{1012} \cdot \sqrt{2} \cdot (i\sqrt{2}) = i \cdot 2^{1012} \cdot \sqrt{2 \cdot 2} = i \cdot 2^{1012} \cdot 2$

$= 2^{1013} \cdot i$

References